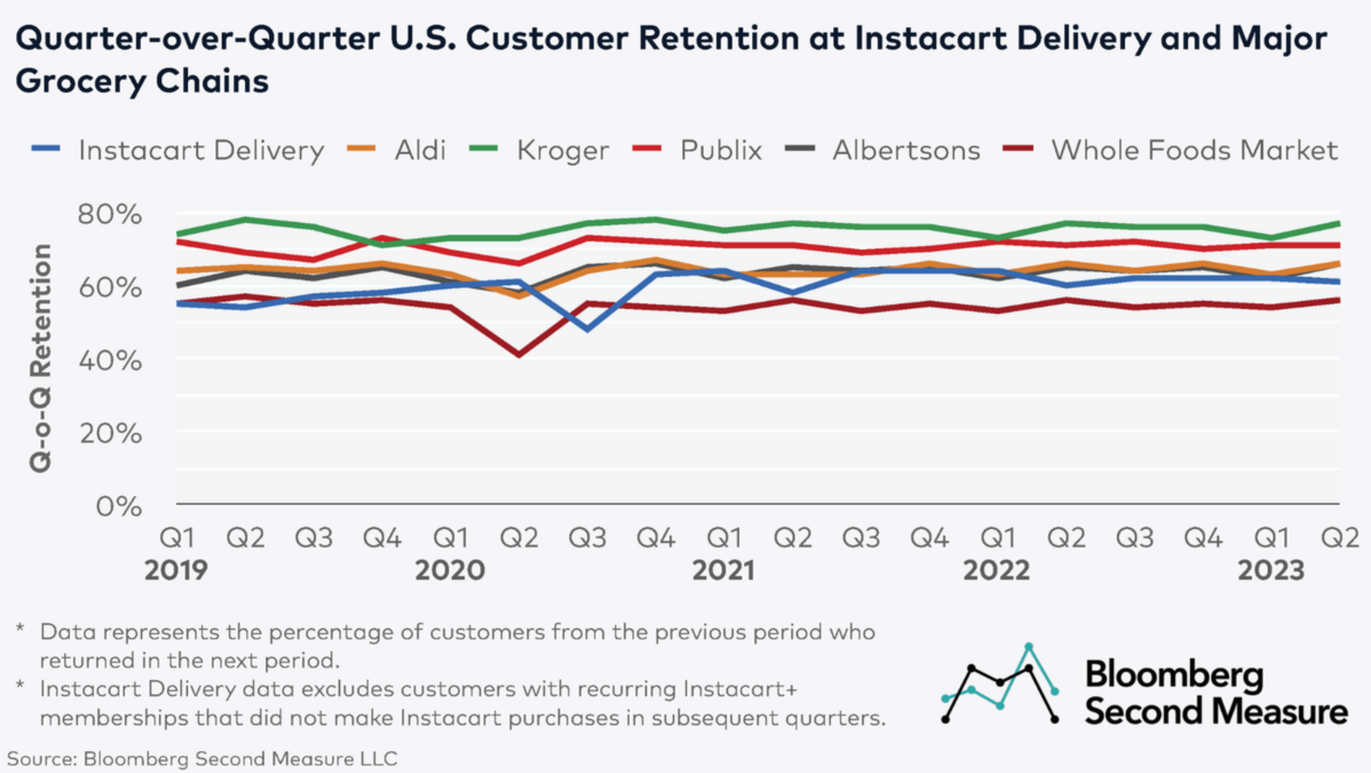
Predicting “Instacart” customer behavior using Machine learning

Machine Learning - DATS 6202

Group 8

Haeyeon Jeong, Sai Rachana Kandikattu, Snehitha Tadapaneni

instacart



- Instacart IPO in Sept 2023 amid slowing growth & rising competition
- Online Grocery sales hit $10.3B in Feb 2025 (↑31% YoY)
- Predicting user purchasing behavior is more crucial than ever

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Project Goals

Improve Customer Retention

- Use machine learning to predict reorders and provide personalized recommendations.

Increase Market Competitiveness

- Provide actionable insights to help Instacart stay ahead of competitors.
- Boost customer loyalty and increase revenue by improving reorder predictions

Optimize with Machine Learning

- Apply feature engineering, sampling strategies, and model tuning for optimized performance.

Methods

Data Preprocessing

Merging CSV Files (6)

Sampling Data

Creating New Features (43)

Multicollinearity check

Hyperparameter Tuning

Models:

- LogisticRegression
- RandomForestClassifier
- XGBClassifier

GridSearchCV

Model Selection

Dataset Overview

1 — aisle_id — 2 — department_id — 3

AISLES

Rows: 134
Columns: 2

PRODUCTS

Rows: 49,688
Columns: 4

DEPARTMENTS

Rows: 21
Columns: 2

6 — order_id — 4,5 — product_id

ORDERS

Rows: 3,421,083
Columns: 7

ORDER_PRODUCTS_PRIOR

Rows: 32,434,489
Columns: 4
(sampling 1M)

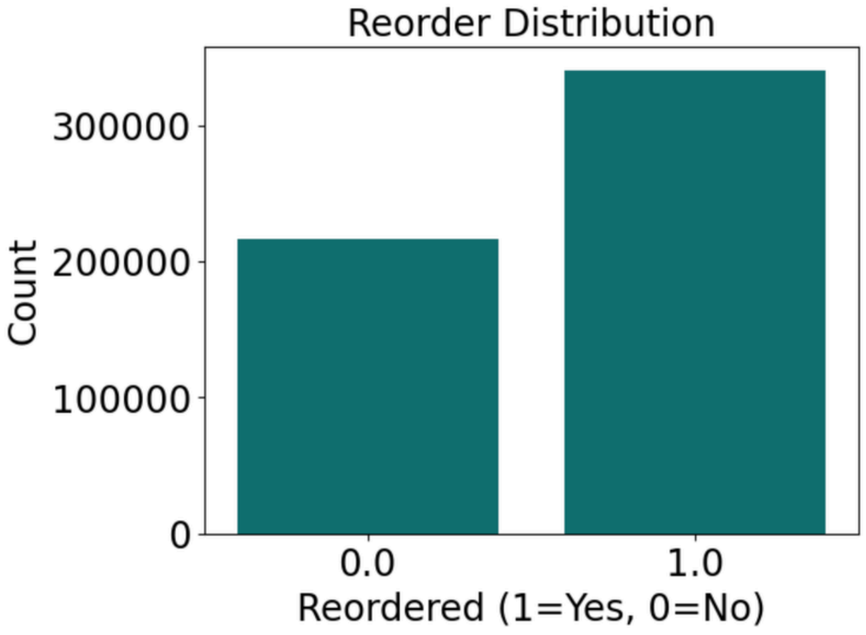
ORDER_PRODUCTS_TRAIN

Rows: 1,384,617
Columns: 4

Data Preprocessing

43 New Features			
User-Product (9)	**User (11)**	**Product (9)**	**Order (6)**
• user_product_cnt • user_total_product_cnt • user_product_dow_mean • user_product_hod_mode • Weekend • user_product_daytype_avg • user_product_atc_mean • user_product_atc_min • user_product_at_max	• orders_total • user_sum_reord • user_reord_prop • user_prod_reord_prop • order_frequency_days • user_uniq_pr • user_mean_order_size • favorite_order_dow • mode_order_hour • weekday_orders_per_day • weekend_orders_per_day	• product_ord_count • product_reord_count • product_reord_count_prop • prod_uniq_us • product_cart_mean • product_dow_mode • product_hour_mode • weekday_prop_norm • weekend_prop_norm	• order_dow_reordered • order_dow_reordered_prop • order_hod_reordered • order_hod_reordered_prop • order_dow_hod_reord_count • order_dow_hod_reord_prop
Cart Order (2)	**Aisle (2)**	**Department (2)**	**Reorder (2)**
• atc_reordered • atc_reordered_prop	• aisle_reordered • aisle_reordered_prop	• dep_reordered • dep_reordered_prop	• ono_dsp_reord • ono_dsp_reord_prop

Exploratory Data Analysis

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Overall reorder rate: 61.12%
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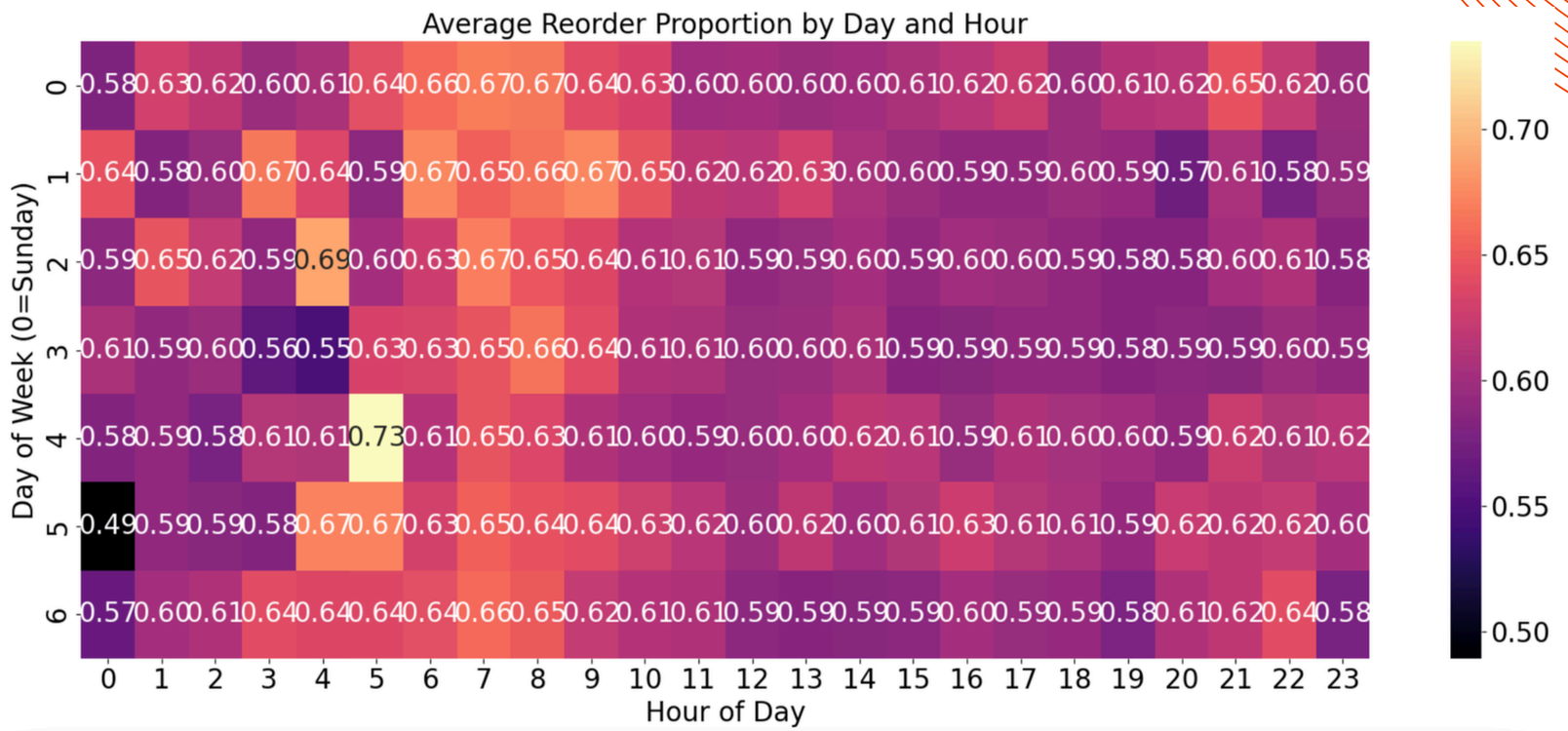
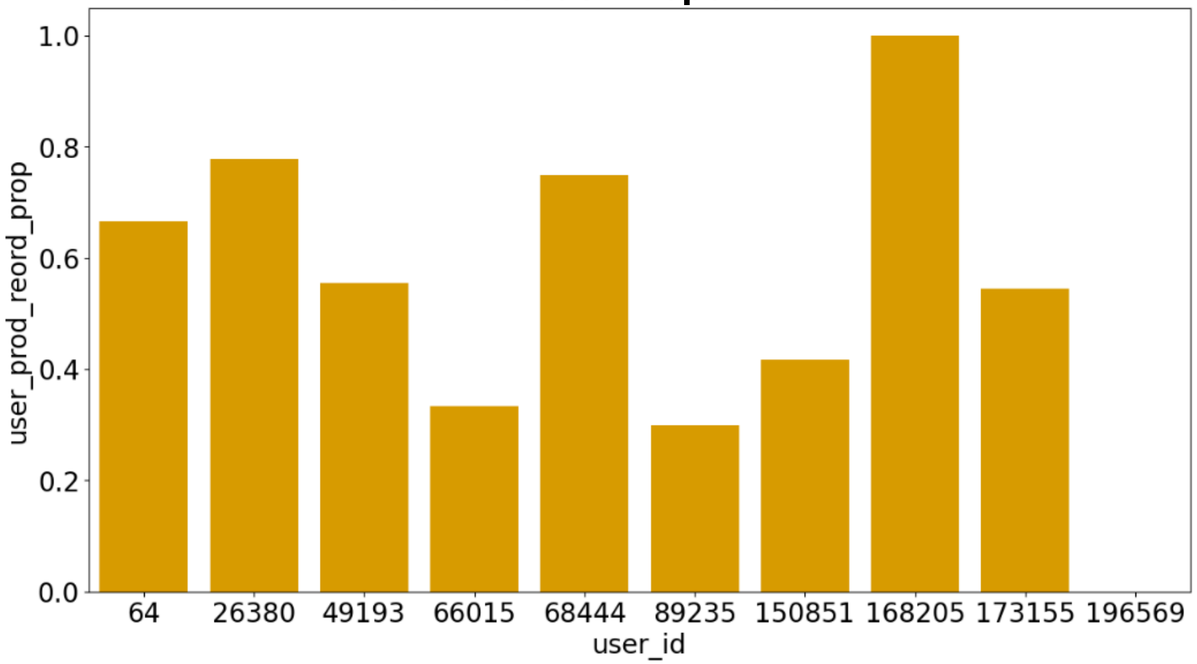


UNDERSTANDING REORDER PATTERNS

Exploratory Data Analysis

User Reorder Proportion



Product Reorder Count Proportion

	product_id	product_name	product_reord_count_prop
27999	40722	Sparkling Citrus Beverage, with Natural Pulp	1.0
556954	28917	Collards	1.0
363364	29478	Fudge Graham Nutrition Bar	1.0
363363	29096	Milk Thistle 150 Mg, v-caps	1.0
363459	27445	Xylitol Peppermint Gum	1.0
96837	33323	Il Mascarpone	1.0
96833	24730	Chomperz BBQ Seaweed Snack	1.0
487898	17729	Ten Calorie Lemon Lime Soda	1.0
435358	41083	Gluten Free Wheat Free Pancake & Waffle Mix	1.0
543606	31093	Original Tofu Pups	1.0

USER AND PRODUCT REORDER BEHAVIOR

Model and Evaluation

01 LOGISTIC REGRESSION

02 RANDOM FOREST CLASSIFIER

03 XGB CLASSIFIER

Hyperparameter Tuning

Logistic Regression	• tol = [10 ** -5, 10 ** -4, 10 ** -3] • C = [1, 10]
Random Forest Classifier	• max_depth = [10, 20] • n_estimators = [50, 100, 200]
XGB Classifier	• max_depth = [3, 5] • n_estimators = [50, 100, 200] • learning_rate = [0.001, 0.1, 0.2]

Model Result

Model	Best Validation F1 Macro Score	Best Parameters	Notes
Logistic Regression	0.8919	{'model__C': 1, 'model__tol': 0.001}	Fast training, good baseline, handles class imbalance well
Random Forest Classifier	0.9011	{'model__max_depth': 20, 'model__n_estimators': 200}	Strong performance, interpretable feature importances
XGB Classifier	**0.9022**	{'model__learning_rate': 0.2, 'model__max_depth': 3, 'model__n_estimators': 50}	Best performance, powerful but may overfit if not tuned carefully

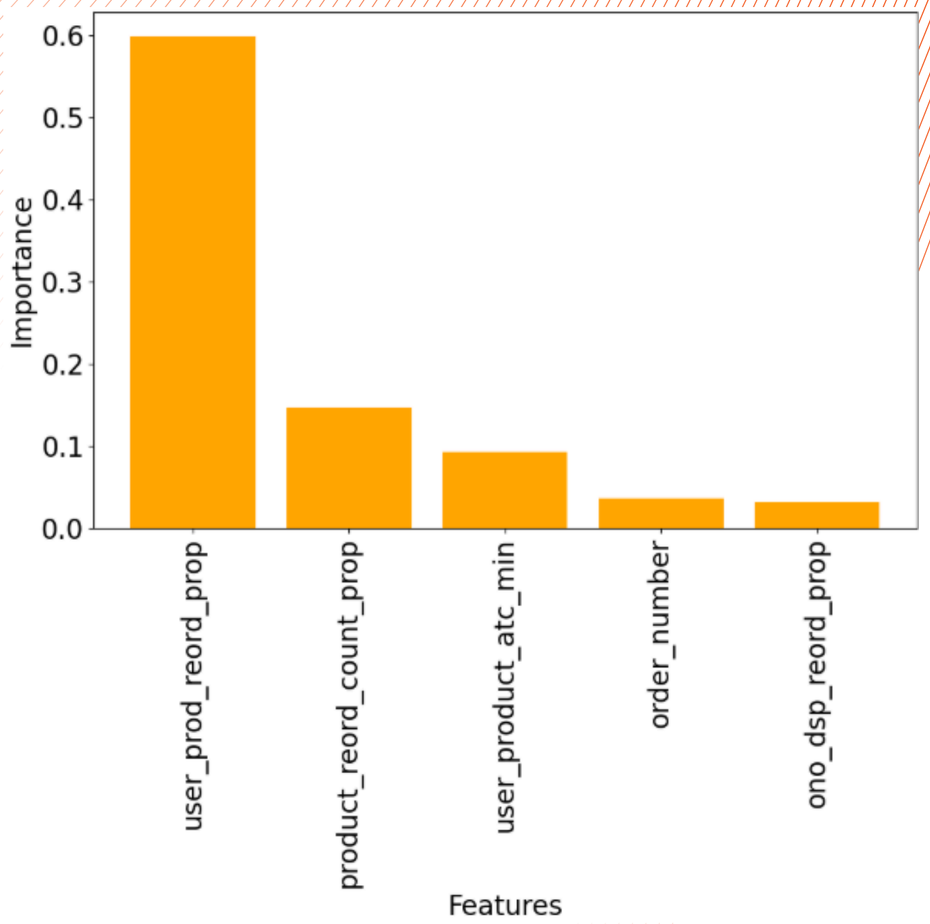
Conclusion

- XGBoost delivered the best performance (**F1 Macro Score: 0.9022**)

- Ensemble methods like XGBoost captured complex patterns better than simpler models

- Feature analysis showed key drivers of reordering behavior:

- User-specific reorder proportion, was the strongest predictor, showing **personal loyalty**.
- Product reorder count proportion, reflected **product popularity**.
- Add-to-cart position, reflected **item priority**.



Thank

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